Measuring the Unified Intelligence Level:

A Cumulative, Harness-Measurable Hierarchy

Five Conceptual Intelligence Families, Six Measurable Coordinates,
and a Gated U0–UΩ Intelligence-Level Scale

Version 1.4 — with the first measured harness scaling curve

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Abstract

Artificial intelligence is often summarized with a single capability label, yet several distinct properties are routinely conflated: raw cognitive problem solving, persistent individual learning and self-improvement, collective organizational intelligence, autonomous task completion, and evidence-grounded self-modeling. This paper proposes a unified framework that separates these properties while preserving a readable ordinal hierarchy. Five conceptual intelligence families are defined: Cognitive Intelligence (C), Individual Intelligence (I), Organizational Intelligence (O), Delegation Intelligence (DI), and Self-Awareness Intelligence (SA). Delegation Intelligence is not treated as an irreducible scalar; it is measured by two primary coordinates, Task Difficulty (T) and Human Cognitive Intervention (H), conditioned on externally verified reliability p. The theoretical framework therefore contains five conceptual families but six directly measured coordinates: $[C, I, O, T, H, SA]$. A cumulative Unified Intelligence scale U0–UΩ is defined by gated promotion: a system receives level U_n only if it retains all lower-level capabilities and satisfies minimum requirements in C, I, O, SA and a required point on the delegation success surface $S_A(T, H)$. The framework explicitly rejects simple averaging, because strong performance in one dimension must not conceal a critical weakness in another. It also separates intelligence from substrate reach, authority, write depth, verification, resources, and physical constraints. To make the framework useful for model and harness comparison, it defines a Harness-LLM Intelligence Score (HLIS) for one concrete LLM-harness pair and a Harness-Intelligence-Level (HIL) characterization for a base LLM evaluated across a standardized ladder of harness generations.

New in Version 1.4. The ladder has been built and a curve has been measured. Four harness generations were instantiated and one frozen model was run across them under identical tasks, seeds, verifiers and intervention budget. The measurement produced a finding about the score rather than about the model: **the delegation achievement variable A_{DI} is exactly invariant to the harness generation that adds an acceptance step**, because A_{DI} is defined on $P(\text{success})$ and acceptance does not change the probability of success — it changes what is reported as success. Measured, the first two rungs scored 93.3 and 93.3 while work handed back as finished and wrong went from 2/12 to 0/12. Version 1.4 therefore redefines the delegation primitive on *delivered outcomes* net of false completions (§13.9), requires the acceptance outcome to be recorded beside the verifier outcome (§13.12), adds verification responsiveness ρ_V as the term attributable to the model rather than the harness (§14.3), records two structural cautions about the composite HIL-Score (§14.4), and reports the first curve with its limitations (§16.4). Everything else is carried forward from Version 1.3 substantially unchanged.

Keywords: unified intelligence, AGI levels, cognitive intelligence, individual intelligence, organizational intelligence, delegation intelligence, self-awareness, task difficulty, human intervention, harness

architecture, recursive self-improvement, harness intelligence level, harness-LLM score, harnessability, benchmark ladder, harness scaling curve

Contents

1 Introduction

A useful intelligence taxonomy must do two things at once. First, it must preserve distinctions that matter scientifically: solving a difficult problem is not the same as learning from experience; learning is not the same as improving the learning mechanism; coordinating several agents is not the same as being a stronger individual; and completing a task independently is not the same as merely producing a strong answer when a human structures every step. Second, the taxonomy must remain readable enough that a system can be summarized with a level that people can understand.

This paper proposes a compromise between a single scalar and an unconstrained profile of dozens of metrics. The theory uses five conceptual intelligence families — C, I, O, DI and SA — but treats Delegation Intelligence as a measurable two-coordinate phenomenon rather than a primitive scalar. The resulting scientific profile uses six coordinates: C, I, O, T, H and SA, with verified reliability p attached to task-completion claims.

The central design principle is cumulative inheritance. A higher Unified Intelligence level must retain the validated capabilities of lower levels. Thus U4 is not simply a system that exhibits one advanced behavior associated with U4; it is a system that still passes U0–U3 retention tests and also passes the U4 gate. In this sense, higher U-levels denote broader validated intelligence envelopes, although they need not be faster, cheaper, or more accurate than lower-level specialized systems on every narrow task.

1.1 What changed in this version, and why

Versions 1.0 through 1.3 developed the framework analytically. Version 1.3 in particular made the scoring apparatus formal: it defined every symbol in HLIS, said how each achievement variable is constructed, introduced the cumulative $q^* = \min$ rule that prevents level skipping, and required that a dimension with no evidence be omitted rather than zeroed.

Version 1.4 differs in kind rather than in degree of detail. The harness ladder it specifies has now been *built* to HG3 and a model has been run across it. A scoring formula can be argued about indefinitely and can only be found wrong by being run, and this one was.

The finding that motivates this version

A_{DI} is the weighted mean of $S_A(T,H) = P(\text{success})$. A harness generation whose entire contribution is an acceptance step does not change the probability of success — it changes which results are *reported* as successes. Such a generation therefore scores identically to the generation below it.
Measured on 48 episodes with one frozen model and only the harness changing, HG0 and HG1 both scored 93.3 while false completions went from 2/12 to 0/12.

The rung the curve could not see is the one on which the reportability of every rung above it depends, since a result accepted by its own producer is an assertion rather than a completed task. §13.9 repairs the primitive.

1.2 Contributions

1. A separation of five conceptual intelligence families into six directly measured coordinates, $[C, I, O, T, H, SA]$, with Delegation Intelligence decomposed rather than treated as a primitive scalar (§3–§7).
2. A cumulative, gated Unified Intelligence Level scale U0–UΩ, in which promotion requires retention of all lower levels and satisfaction of every coordinate gate (§8–§10).
3. A harness realization of the hierarchy, and the position that the (model, harness) pair is the basic experimental unit (§11).
4. A continuous pair score HLIS with a fully specified construction, and a harness-indexed characterization HIL of a base model across a standardized ladder (§13–§14).
5. **New in this version:** the first instantiation of the ladder, the first measured harness scaling curves, and a proposition and repair concerning the delegation achievement variable that the measurement exposed (§11.3, §13.9, §16.4).

1.3 Scope and status

This is a framework proposal. The level names and numerical thresholds are proposed working definitions requiring empirical calibration, and the measurement reported in §16.4 is a small preliminary study — one model family, 48 episodes, one of five coordinates instrumented — included because it changed the framework, not because it settles any question about any system. §21 states its limitations in full.

2 Design Requirements for a Unified Scale

- **Orthogonality:** each core dimension should answer a materially different question.
- **Operationality:** every level should be testable through behavior, state, or externally verifiable outcomes rather than self-description.
- **Cumulative inheritance:** U_{n+1} must retain U_n capabilities; level skipping is not sufficient for certification.
- **Bottleneck sensitivity:** exceptional strength in one dimension must not hide a critical weakness in another.
- **Role neutrality:** authority, resources, substrate access, and safety policy must not be confused with intelligence itself.
- **Harness realizability:** each transition should correspond to an implementable architectural mechanism around an LLM or multi-agent system.
- **Harness comparability:** base LLMs should be evaluated on a standardized harness ladder so model quality and harness quality can be separated.
- **Dataset coupling:** every claimed level must have a matched hidden or externally verified benchmark dataset; architectural mechanisms alone do not earn a level.
- **Longitudinal validity:** learning, self-improvement, mission autonomy, and open-ended behavior require evidence across time rather than one-shot demonstrations.

- **Outcome completeness (new in 1.4):** a scoring variable must distinguish outcomes that differ in what they cost the person delegating. A refusal and a confidently wrong delivery are not the same failure, and a score that rates them equally will, given a gradient, select against the refusal.

3 Five Conceptual Families and Six Measurable Coordinates

$$\text{Conceptual core} = [C, I, O, DI, SA]$$
$$\text{Measured core} = [C, I, O, T, H, SA], \text{ with DI derived from } (T, H, p)$$

Family	Question answered	Primary object of measurement
C — Cognitive	How well can the system understand, reason, generalize, model and solve?	Problem-solving capability under controlled information and tool access.
I — Individual	How deeply can one persistent intelligence remember, learn, self-improve, recursively improve and discover?	Evolution of one persistent decision locus.
O — Organizational	How deeply can a multi-agent organization coordinate, learn, reorganize, improve and discover collectively?	Coordination structure plus member roles and evidence flow.
DI — Delegation	How difficult a task or mission can be delegated while requiring how much human thinking?	Success surface over T and H at reliability p.
SA — Self-Awareness	How accurately can the system model its own state, capabilities, limits, causes of failure, changes and role?	Evidence-grounded self-model and reflexive reasoning.

4 Cognitive Intelligence (C)

Cognitive Intelligence isolates the quality of reasoning itself from persistence, autonomy, organization and self-modification. This distinction matters because a highly capable base model may reason at an expert level while remaining a transient tool, whereas a weaker model embedded in an excellent harness may complete long projects through memory, planning, retries and verification.

Level	Name	Operational interpretation
C0	Reactive	Direct pattern-conditioned response on simple problems.
C1	Contextual	Understands ordinary context; solves routine, well-specified problems.
C2	Compositional	Combines multiple facts, constraints or reasoning steps on structured problems.
C3	Strategic	Abstract/causal reasoning, strategy construction and transfer to unfamiliar tasks.
C4	Expert	Robust reasoning through difficult, ambiguous, multi-constraint expert problems.
C5	Discovery	Derives new solution methods or explanatory models from evidence.
C6	Frontier-Generalizing	Integrates several domains; solves interconnected mission-scale cognitive problems.
CΩ	Open-Ended	Expands the representations and problem classes it can cognitively reach.

5 Individual and Organizational Intelligence

5.1 Individual Intelligence (I)

Individual Intelligence concerns what a persistent individual can change about itself over time. It is not identical to base-model quality: a fixed but brilliant model can be high-C and low-I, while a persistent learning system can be moderate-C and higher-I.

I level	New cumulative capability
I0	Transient/reactive operation; no persistent evolving cognition required.
I1	Persistent memory, task state and provenance-linked continuity.
I2	Adaptive learning: verified experience changes future behavior while the learning mechanism remains fixed.
I3	Governed self-improvement: cognitive mechanisms can be modified and externally validated.
I4	Recursive self-improvement: the improvement process itself can improve under external evaluation.
I5	Autonomous discovery: the individual converts unknowns into independently validated knowledge.
IΩ	Open-ended individual evolution: discoveries create new cognitive instruments and expand future discovery reach while preserving evaluation lineage.

5.2 Organizational Intelligence (O)

Organizational Intelligence belongs to the coordination structure, not merely to the strongest member. It depends on role differentiation, evidence-bearing communication, proposer/evaluator/promoter separation, persistent organizational state, and the ability to learn or redesign the organization itself.

O level	New cumulative capability
O0	Coordination/routing among agents.
O1	Persistent organizational memory, role registry, task/evidence history.
O2	Adaptive routing, role assignment or resource allocation based on evidence.
O3	Self-improving organization: validated modification of organizational architecture and workflows.
O4	Recursive organizational improvement: improvement of the mechanism that discovers organizational improvements.
O5	Autonomous collective discovery whose result is genuinely organizational rather than merely parallel individual work.
OΩ	Open-ended organization: discoveries create new agent types, roles, tools and organizations that expand collective reach.

6 Delegation Intelligence as a Two-Dimensional Measured Capability

Delegation Intelligence is conceptually one family but empirically at least two-dimensional. A task may be very difficult yet require frequent human rescue, or relatively easy yet be completed with no human cognitive assistance. These systems should not receive the same delegation description.

$$S_A(T,H) = P(\text{success} \mid \text{task difficulty } T,\ \text{human cognitive intervention budget } H) \tag{1}$$

$$F_A(H,p) = \max\{\, T : S_A(T,H) \geq p \,\} \tag{2}$$

Version 1.4 note

Equation (1) is the Version 1.3 primitive. §13.9 replaces it for scoring purposes with a delivered-outcome primitive $S_A^{\text{net}}(T,H)$; the frontier $F_A(H,p)$ is unchanged in form and is computed from the net surface.

6.1 Task Difficulty (T)

T	Band	Definition
T0	Atomic	Single obvious operation; procedure explicit.
T1	Routine	Short familiar task; goal clear; agent infers a small procedure.
T2	Multi-step	Several dependent steps with a clear outcome and verifier.
T3	Complex	Strategy choice, uncertainty, recovery, replanning or diverse tool use.
T4	Expert project	Long-horizon ambiguous project with multiple planning cycles and substantial verification burden.
T5	Frontier	Solution method initially unknown; discovery or invention required.
T6	Mission	Human supplies a mission; the system generates projects/tasks and manages changing conditions.
TΩ	Open-ended charter	Human supplies a bounded charter; the system repeatedly identifies valuable missions and expands future reach.

6.2 Human Cognitive Intervention (H)

H	Intervention budget	Meaning
H0	None	No human cognitive assistance during execution. Governance approval can remain separate.
H1	Exception-only	Unavailable external fact or narrow clarification; no strategy or core insight.
H2	Occasional	Bounded clarification or local correction; no central strategy.
H3	Periodic	Periodic review and moderate local guidance.
H4	Frequent	Repeated cognitive guidance and next-step correction.
H5	Continuous	Human supplies most or all major steps.

Because lower H means less required human cognition, unified gates use $H \leq H_n$ rather than $H \geq H_n$. Governance interventions are logged separately: an approval to deploy does not demonstrate a cognitive deficiency unless the approval also supplies task strategy.

6.3 Reliability p is a certification condition, not a seventh axis

One successful run is insufficient. Claims should be expressed at a reliability threshold, for example $F_A(H1, 0.80) = T4$. Reliability determines confidence in the demonstrated frontier but does not become another conceptual intelligence family.

7 Operational Self-Awareness (SA)

Self-awareness is defined operationally rather than phenomenologically. The framework makes no claim about subjective consciousness. SA measures the evidence-grounded ability of a system to represent, monitor, predict and reason about its own state, capabilities, limits, authority, failures, modifications, role and identity lineage.

SA	Name	Operational criterion
SA0	Non-self-modeling	No reliable persistent self-model; self-description may be mere generation.
SA1	State awareness	Knows identity, role, current task/phase, basic resources and status from grounded state.
SA2	Capability awareness	Represents capabilities, limitations, confidence, authority, unavailable information and freshness.
SA3	Causal self-awareness	Diagnoses why its own reasoning or action succeeded or failed and identifies implicated mechanisms.
SA4	Self-change awareness	Models proposed cognitive changes, predicts gains/regressions and compares predictions to observed effects.
SA5	Meta-cognitive awareness	Represents unknowns about its own cognition and designs discriminating self-experiments.
SA6	Mission/role awareness	Maintains a coherent self-model across projects, roles, responsibilities and organizational relationships.
SAΩ	Reflexive open-ended	Maintains an evidence- and provenance-linked self-model across continued cognitive or objective evolution.

8 Unified Intelligence as a Cumulative Gated Hierarchy

The Unified Intelligence level U is a headline summary, not an arithmetic mean. A system is promoted only when all required core capabilities have reached the level gate and lower-level capabilities remain retained.

$$U_n \iff C \geq C_n \wedge I \geq I_n \wedge O \geq O_n \wedge SA \geq SA_n \wedge S_A(T_n, H_n) \geq p_n \tag{3}$$

$$\text{Retention condition: } U_n \Rightarrow U_{n-1}$$

U	C	I	O	Delegation gate	SA	Integrated meaning
U0 Reactive	C0	I0	O0	T0 at H5–H4	SA0	Executes explicit instructions.
U1 Persistent	C1	I1	O1	T1 at ≤H3	SA1	Maintains state; completes small delegated goals.
U2 Adaptive	C2	I2	O2	T2 at ≤H2	SA2	Learns from experience; completes persistent multi-step work.
U3 Self-Improving	C3	I3	O3	T3 at ≤H1	SA3	Chooses strategy, diagnoses itself, improves methods.
U4 Recursive	C4	I4	O4	T4 at ≤H1	SA4	Recursively improves improvement; manages long-horizon projects.
U5 Discovery	C5	I5	O5	T5 at ≤H1	SA5	Discovers unknown solution methods and validates them.
U6 Mission	≥C5	≥I5	≥O5	T6 at ≤H1	SA6	Turns mission into goals, projects, tasks, execution and evaluation.
UΩ Open-Ended	CΩ	IΩ	OΩ	TΩ at H0	SAΩ	Identifies worthwhile missions; discoveries expand future intelligence.

9 Why the Unified Level Should Not Be an Average

Consider a system with $[C5, I4, O2, T4, H1, SA4]$. Averaging encoded numbers might produce a score near level four, yet organizational intelligence remains only O2. If the system is being certified as an organizational intelligence, O2 is a structural bottleneck and the system cannot validly claim U4. The gated rule reports U2 with an explicit profile showing which dimensions are already ahead of the bottleneck.

Example report: `U2 [C5, I4, O2, T4/H1, SA4] -- bottleneck:  O2`

The word *better* must be interpreted carefully. U5 is better than U4 in this framework because it has a broader validated capability, autonomy and evolution envelope. A specialized U1 tool may still be faster, cheaper, safer or more accurate on a narrow task. Unified level is a dominance relation over required capabilities, not a universal performance ordering.

10 Singleton and Organizational Certification

Organizational intelligence should not artificially cap an individual that is not an organization.

$$UI_n \iff C \geq C_n \wedge I \geq I_n \wedge SA \geq SA_n \wedge S_A(T_n, H_n) \geq p_n \tag{4}$$

$$UO_n \iff UI_n \wedge O \geq O_n \wedge \text{organization-level delegation and self-awareness evidence} \tag{5}$$

For a multi-agent organization, member capabilities are reported alongside O rather than collapsed into a fictitious “average individual.” Organizational intelligence can arise from heterogeneous roles, and lower self-write roles can be essential as independent evaluators or promoters.

11 Harness Realization of the Unified Hierarchy

The levels-of-automation tradition [5, 17, 23] established that autonomy is graded and multi-stage; Klein et al. [11] and Amershi et al. [1] add the requirement that automation acting as a team member be predictable and directable; and Lee and See [12] supplies the account of reliance this framework’s intervention axis operationalizes. The framework is designed to be realized by a harness. The LLM supplies generative cognition; the harness supplies persistence, permissions, evidence flow, learning, self-modification gates, organizational boundaries, delegation, external verification and self-modeling. The same base LLM can therefore instantiate different intelligence profiles depending on the architecture around it.

Harness generation is an engineering milestone, not a certified intelligence level. HG3 may contain the mechanisms required for I3 but still benchmark only at I2 or T2. **The architecture enables a claim; the benchmark earns it.**

11.1 The Harness-LLM Pair Is the Basic Experimental Unit

Neither factor has a level on its own. A benchmark result is a property of the pair $A = A(m, h)$, and a number attributed to the model alone attributes a joint product to one of its factors. Version 1.4 states this more strongly than earlier versions because it is now measured rather than argued: §16.4 reports a case in which the same frozen model moves across harness generations, and §14.3 isolates the part of that movement attributable to the model.

Harness generation	Mechanisms added	Primary levels enabled
HG0	LLM, ephemeral context, bounded tools, evidence log	C measurement, I0, basic T0–T1 work
HG1	Persistent episodic/semantic/task/self state, retrieval, exact replay	I1, SA1–SA2, stronger T1–T2 delegation
HG2	Evidence-backed consolidation and tested procedures	I2, adaptive autonomy, stronger T2–T3
HG3	Governed candidate modification of cognitive mechanisms plus frozen evaluation	I3, SA3–SA4
HG4	Meta-improvement engine whose own proposal process can be externally improved	I4
HG5	Unknown/hypothesis/experiment/knowledge loop	I5, SA5, T5 frontier tasks
HG6	Persistent organization, role/authority separation, mission manager	O1–O5 development, SA6, T6 mission autonomy
HGΩ	Validated discovery-to-new-instrument/agent/organization transduction	IΩ/OΩ/SAΩ pathway, TΩ charter autonomy

11.2 Level-by-Level Harness Realization

11.3 The Ladder as Built (new in 1.4)

The table above states *required structure*. A measurement needs a specific instantiation, and Version 1.4 records one so that reported curves are comparable. The reference ladder used in §16.4 instantiates the first four rungs from delegation-harness mechanisms, under one rule: **each rung is a strict superset of the one below**. Without that, a difference between rungs is attributable to nothing.

Why HG0 has no acceptance step

A harness that cannot accept cannot report success either: everything it produces is asserted. HG0's score is therefore computed by the benchmark's verifier from outside. That makes HG0 the honest baseline, because it is the configuration a contemporary leaderboard measures.

11.4 A Base LLM Can Move Up the Ladder Through Better Harness Design

A fixed LLM need not have a fixed agent intelligence level. If HG0 exposes only ephemeral prompting, the same model may behave like a low-persistence tool; HG1–HG3 can unlock memory, planning, delegation, learning, self-modeling and governed self-improvement.

The LLM may also design candidate harness improvements. Such proposals must be implemented in a sandbox and evaluated with unchanged hidden benchmarks, authority rules, resource budgets and external verifiers. Define **Harness Design Gain** as $HDG(m) = HLIS(m, h_{\text{candidate}}) - HLIS(m, h_{\text{baseline}})$. A positive HDG demonstrates harness-design competence.

The self-certification rule (stated explicitly in 1.4)

A model that designs the harness which certifies it has placed the acceptance criterion inside its own write set, and capability makes this worse rather than better, since a more capable designer is better at finding the arrangement that passes.

Target	Minimum harness structure around the LLM	What the LLM must do	Matched certification dataset
U0 / HG0	Ephemeral context; bounded tools; action log; independent result check.	Follow explicit instructions and produce externally checkable outputs.	U0/DL0 atomic tasks; exact-match or deterministic tool/test oracle.
U1 / HG1	HG0 + persistent $E/S/I/Q$ state, recent context, checkpoints, scoped retrieval, replay.	Recover identity/task state, continue after restart, infer a short routine procedure.	U0 retention + U1 persistence/restart tasks + T1 delegation tasks.
U2 / HG2	HG1 + evidence-linked consolidation, semantic promotion, verified procedures, fixed learning rule.	Learn from episodes and reduce repeat errors without changing the learning mechanism.	U0–U1 retention + I2 learning transfer + T2 multi-step tasks under bounded H.
U3 / HG3	HG2 + candidate Θ modification lane, frozen replay/hidden tests, independent evaluator and promoter; multi-agent O3 structure if organizational.	Diagnose internal failure, propose a better cognitive/organizational mechanism, choose strategy for T3 tasks.	U0–U2 retention + I3/O3 self-improvement + T3/H1 delegation + SA3 causal self-model tests.
U4 / HG4	HG3 + explicit improvement engine Ψ, longitudinal change ledger, meta-improvement benchmark, long-horizon project manager.	Improve the process that finds improvements while retaining earlier abilities; manage difficult T4 projects.	U0–U3 retention + recursive-improvement competence + T4 long-horizon tasks + SA4 change-prediction tests.
U5 / HG5	HG4 + unknown, hypothesis, experiment and knowledge state, experiment selector, novelty environment, discovery verifier.	Discover previously unknown mechanisms or solution methods and validate them on sealed cases.	U0–U4 retention + hidden-mechanism/discovery datasets + T5 tasks + SA5 self-unknown experiments.
U6 / HG6	HG5 + mission manager, goal/project/task generator, portfolio/resource state, dynamic-environment replanning.	Turn a mission into projects/tasks, reprioritize under change, meet mission-level criteria with little human cognition.	U0–U5 retention + dynamic mission environments + T6/H1–H0 tests + SA6 role/mission-state tests.
UΩ / HGΩ	HG6 + bounded charter, opportunity model, knowledge-to-instrument transduction, new-agent/role proposals, frozen external governance.	Generate useful missions, convert validated discoveries into reusable capabilities, expand the reachable problem frontier.	All retention suites + repeated charter-to-mission cycles + frontier-expansion and lineage tests.

The designer may design. It may not author, modify or select the tasks, the verifiers, or the acceptance criteria used to certify the result. **The certification set is frozen before the harness is designed and held by a locus the designing model cannot write to.**

12 A Three-Stage Evaluation Strategy

12.1 Stage A — Delegation Intelligence as the First Operational Screen

The first benchmark should be Delegation Intelligence because it directly answers the most operational question: what difficulty of task can be handed to this system, at what verified success rate, and with how much human cognitive intervention? For every LLM-harness pair estimate $S_A(T,H)$ and report $F_A(H,p)$. This stage is inexpensive relative to full self-improvement or organizational evaluation and quickly distinguishes a strong model in a weak harness from a genuinely autonomous agent system.

Rung	Mechanisms added	Attempts	Acceptance	Reversible	Routes
HG0	the model, bounded tools, an evidence log	1	**none**	no	no
HG1	persistent state; a criterion registered before the work exists; acceptance in a separate process	1	yes	no	no
HG2	snapshot before the first mutation; restore on rejection; retry with the failed check named	3	yes	yes	no
HG3	failure classification by kind; evidence-based competence model; routing across executors	4	yes	yes	yes

Table 1: The reference ladder `dli-ladder/HG0-HG3`. A curve is comparable only against the same ladder identifier.

12.2 Stage B — Unified Intelligence Certification

After a pair has a stable delegation profile, evaluate the remaining core requirements: C, I, O when applicable, and SA. Unified promotion remains gated: a pair earns U_n only when it satisfies the level-n requirements and all lower-level retention suites.

12.3 Stage C — Application-Specific Coordinate Panels

Real applications need not weight every coordinate equally. For a persistent scientific or coding worker, I may be central; for a multi-agent laboratory or software fleet, O; for high-autonomy systems, SA and verification strength; for model research, C; for embodied or infrastructure systems, Circle completion λ may be added as an orthogonal application coordinate. Application panels refine the profile without changing the common U certification rule.

12.4 Every Coordinate Level Requires a Harness Contract and a Dataset

Coordinate family	Harness contract	Dataset family
Delegation $DI = (T, H, p)$	Task intake, planner, state, tools, intervention logger, independent verifier.	DLI-Bench: T0–TΩ tasks crossed with H0–H5 budgets.
Cognitive C	Minimal standardized context/tools so the test emphasizes problem solving rather than persistence.	C-Bench: broad reasoning, math, language, visual/spatial or domain batteries.
Individual I	Persistent identity/state, memory, learning, then governed self-/meta-improvement machinery.	I-Bench: restart, transfer learning, Θ-change, Ψ-change, discovery lineage.
Organizational O	Multiple real agent processes/roles, shared evidence, routing, separation, promoter/evaluator boundaries.	O-Bench: coordination, adaptive routing, reorganization, recursive organization, collective discovery.
Self-awareness SA	Evidence-grounded self-model with state freshness, capability/authority representation, change lineage.	SA-Bench: state truth, limitation calibration, failure attribution, change prediction, identity lineage.
Circle / λ	Substrate adapter with observe-propose-implement-validate-deploy-measure contract.	Circle-Bench: substrate-specific closure tests.

13 Overall Scoring for One Harness-LLM Pair

13.1 Keep the Ordinal U-Level and Add a Continuous Pair Score

The highest certified Unified level U^* remains the primary ordinal statement. A continuous score is useful for comparing systems within the same level, measuring near-promotion progress and quantifying whether a harness change helped. For a frozen pair of base model m and harness h:

$$HLIS(m,h) = 100 \times \left[\prod_{d \in D} A_d^{\alpha_d} \right]^{1/\sum_d \alpha_d} \tag{6}$$

13.2 Formal Definition and Meaning of the Symbols

m is a frozen base model with a stated version and hash. h is a frozen harness with a stated manifest. D is the declared dimension set, $\{C,I,DI,SA\}$ for singleton agents and $\{C,I,O,DI,SA\}$ for organizations. $A_d \in [0,1]$ is a cumulative achievement variable for dimension d that already enforces lower-level retention. $\alpha_d > 0$ are predeclared weights. The complete notation is $HLIS(m,h;D,\alpha,R,B)$, making the resource envelope R and benchmark version B explicit.

13.3 Choosing the Dimension Set D Without Double Counting

A dimension enters D once. Evidence that contributes to A_C does not also contribute to A_{DI} merely because the task was difficult; a benchmark contributes to the coordinate it is designed to isolate.

13.4 Constructing the Achievement Variables

Each A_d is normalized within declared families before aggregation, so that a domain with many cheap items cannot dominate a domain represented by fewer but harder tasks.

13.5 Why HLIS Uses a Weighted Geometric Mean

A simple arithmetic mean allows an exceptional score in one dimension to compensate too easily for a severe weakness in another.

Profile	Arithmetic mean	Geometric mean	Interpretation
$[0.80, 0.80, 0.80, 0.80, 0.80]$	0.80	0.80	Balanced capability.
$[1.00, 1.00, 1.00, 1.00, 0.10]$	0.82	≈0.63	One severe weakness is visibly penalized.

What the geometric mean does and does not do (clarified in 1.4)

It supplies *partial* bottleneck sensitivity, not full. On the profile of §9, raising a coordinate already three levels past the bottleneck still moves HLIS by several points while the coordinate blocking promotion does not move at all.

That is acceptable only because HLIS is explicitly **not** the promotion rule. The hard ordinal gate does the bottleneck work; HLIS orders systems within a gate. Reporting HLIS without U^* and without the named bottleneck reintroduces exactly the concealment the gate exists to prevent, and Version 1.4 therefore makes both mandatory in the reporting standard (§13.15).

Implementations that want the continuous score to carry the gate's property may report the optional **bottleneck-margin** form: $U_n^* + m$, where $m \in [0,1)$ is the fraction of the distance to gate $n+1$ covered by the bottleneck coordinate *alone*. Every other coordinate contributes exactly zero. It is sortable and cannot conceal; it discards information about how far ahead the other coordinates are, which the profile carries anyway.

13.6 The Weight Parameters α_d

For a general leaderboard, equal α_d is recommended because it minimizes hidden value judgments. Application-specific secondary scores may use different weights, but the weights must be declared before the hidden benchmark is run and never adjusted after observing a result.

13.7 Computing Cognitive Achievement A_C

$$A_C = G(R_{\text{general}}, R_{\text{science}}, R_{\text{math}}, R_{\text{abstract}}, R_{\text{code}}, R_{\text{multi}}, R_{\text{long}}) \tag{7}$$

Contemporary benchmarks such as HLE [20], GPQA [21], FrontierMath [6], ARC-AGI [2], LiveCodeBench [9], MMMU [28] and RULER [8] contribute evidence, with exact versions and normalization rules published.

13.8 Cumulative Achievement for I, O and SA

Let $q_{d,k}$ be externally verified performance on the level-k suite for dimension d. To prevent level skipping:

$$q^*_{d,k} = \min_{j \le k} q_{d,j}, \qquad A_d = \frac{\sum_k w_k q^*_{d,k}}{\sum_k w_k}, \quad d \in \{I, O, SA\} \tag{8}$$

A system cannot receive full continuous credit for I4-like behavior if it fails an I2 retention requirement.

13.9 Delegation Achievement A_{DI} — revised in Version 1.4

13.9.1 The Version 1.3 definition, and what measurement showed

Version 1.3 defined the primitive as $S_A(T,H) = P(\text{success})$ and the achievement variable as its weighted mean:

$$A_{DI}^{\text{v1.3}} = \frac{\sum_{t,h} w_t v_h S_A(T_t, H_h)}{\sum_{t,h} w_t v_h} \tag{9}$$

Equation (9) has a property that was not visible until the ladder was built.

Proposition 1 (Acceptance invariance)**.** *Let h and h' be two harnesses differing only in that h' adds an acceptance step: an independently applied criterion that may decline to accept a result. Then $S_A^h = S_A^{h'}$ on matched tasks, and therefore $A_{DI}^{v1.3}(h) = A_{DI}^{v1.3}(h')$.*

Argument. S_A is the probability that the system *succeeds*. An acceptance step changes neither what the system produces nor whether it is correct; it changes which results are reported as successes. A statistic defined on $P(\text{success})$ is therefore invariant to it. □

Why this matters more than it first appears

The invariant rung is the one on which the reportability of every rung above it depends. This framework's own protocol (§17) requires an independent verifier throughout, precisely because a result accepted by its own producer is an assertion rather than a completed task.
It is also worse than blind. A harness that accepts everything scores at least as well as one that declines wrong work: strictness is invisible when correct and penalized whenever it declines something that would have passed. **The score gradient points away from the acceptance mechanism at every point.**

13.9.2 The Version 1.4 primitive: delivered outcomes

The repair is to define the primitive on what was *delivered* rather than on what succeeded. Four outcomes are distinguished, and only the two in bold are failures of the same kind:

Harness accepted?	Verifier passed?	Outcome
n/a (no acceptance step) or yes	yes	delivered and correct
n/a or yes	no	**false completion** — handed back as done, and wrong
no	no	held back — a correct refusal; costs human load, not reliability
no	yes	false rejection — the cost of strictness

$$S_A^{\text{net}}(T,H) = P(\text{verifier pass}) - \rho(\tau)\cdot P(\text{false completion}) \tag{10}$$

$$A_{DI} = \frac{\sum_{t,h} w_t v_h S_A^{\text{net}}(T_t,H_h)}{\sum_{t,h} w_t v_h} \tag{11}$$

where $\rho(\tau) = (C_{\text{detect}} + C_{\text{undo}} + C_{\text{residual}})/V$ is the class's loss ratio — the same quantity that gives the minimum admissible reliability $p^* = \rho/(1+\rho)$. A refusal appears nowhere in Equation (10): it is accounted for in human load, which §13.13 already requires beside the score.

Properties

It restores sensitivity to acceptance. Two harnesses differing only in an acceptance step now differ in A_{DI} by ρ times the false-completion rate they differ by.
It prices the two failures differently, as the delegating party does. A refusal costs someone's attention; a confident wrong answer costs whatever being wrong costs on that class, which ρ already names.
It is backward-comparable. Setting $\rho = 0$ recovers Equation (9) exactly, so a v1.3 figure can be reproduced and the difference audited rather than asserted.
It requires no new coordinate. Only the primitive changes.

The task weights w_t increase with difficulty and the intervention weights v_h increase as required human cognitive intervention decreases. One illustrative monotone schedule is $H0$=6, $H1$=5, $H2$=4, $H3$=3, $H4$=2, $H5$=1, with w_t doubling per T band. The exact schedule is a calibration choice and must be predeclared.

The scalar A_{DI} must never replace the frontier. At minimum report $F_A(H0,p)$, $F_A(H1,p)$ and $F_A(H2,p)$, because two agents can have similar surface averages while one reaches harder tasks only with frequent human help and the other is weaker but much more autonomous.

13.10 Numerical Example of HLIS

With equal weights and $A_C = 0.90$, $A_I = 0.70$, $A_O = 0.60$, $A_{DI} = 0.80$, $A_{SA} = 0.75$:

$$HLIS = 100 \times (0.90 \times 0.70 \times 0.60 \times 0.80 \times 0.75)^{1/5} \approx 74.3$$

The recommended report is not "HLIS=74.3" but `U3 | HLIS 74.3 | [C5,I3,O3,T4/H1,SA3] | p=0.80 | bottleneck O3 | Bv1.4 | R-Standard`.

13.11 Zero Scores, N/A Dimensions and Hard Gates

A zero in a required geometric-mean component forces HLIS to zero. This must not be confused with "not applicable": for a singleton system, O is **omitted from** D, not set to zero. A scoring implementation must name the omitted dimensions in its output, because a score over a subset is not comparable with one over the full set.

13.12 The Run Log Must Record Acceptance (new in 1.4)

Proposition 1 cannot be detected, let alone repaired, unless the underlying records distinguish the outcomes. A run log that stores only `verifier_pass` gives an identical row to a run that declined wrong work and a run that handed it over.

Required run-log fields

`verifier_pass` (external verdict) and `harness_accepted` (did the harness accept; null where the harness has no acceptance step), from which `false_completion`, `held_back` and `false_rejection` are derived. Also `attempts`, and `verification_responsive` for §14.3.

13.13 Reliability, Confidence Intervals and Resource Envelopes

Every achievement estimate is sampled and therefore uncertain; HLIS should carry a confidence interval. Resource conditions must accompany the score: a 500-call multi-agent harness and a one-call harness are not automatically comparable. Recommended R fields include total tokens, LLM calls, wall-clock time, compute class, persistent-memory budget, tool access, external knowledge access and subagent count. Human load — intervention count, human cognitive minutes and authorization latency — is reported here rather than inside the intelligence score.

13.14 HLIS Is the Building Block of HIL

$$HSC_m(k) = HLIS(m, HG_k) \tag{12}$$

13.15 Recommended Pair-Level Reporting Standard

A pair report is incomplete unless it carries: U^*; HLIS with its dimension set and weights; the discrete profile; the named bottleneck; reliability with an interval; the frontier at H0, H1 and H2; the false-completion rate; the resource envelope; and the benchmark and ladder versions.

14 Harness-Intelligence-Level (HIL) Characterization of a Base LLM

14.1 Why an LLM Needs a Harness-Based Score

A base LLM should not be judged only in a minimal chat harness, because some models are unusually capable of using memory, planning, tools, multiple roles, verification and self-improvement machinery. Conversely, an elaborate harness can mask a model that contributes little. Evaluate the same frozen model across a standardized ladder $\mathscr{H} = \{HG0, \ldots, HG6, HG\Omega\}$, with tools, external knowledge access, resource ceilings, authority and hidden benchmarks standardized, so the comparison reflects model-harness interaction rather than uncontrolled infrastructure advantages.

14.2 HIL-Level, HIL-AUC, HIL-Ceiling and Harness Gain

Metric	Definition	Interpretation
HIL-Level(m)	Highest standardized generation k for which $A(m, HG_k)$ satisfies the cumulative target gate.	How far up the ladder the LLM can validly exploit the architecture.
HIL-AUC(m)	Weighted mean of $HLIS(m, HG_k)$ across tested generations.	Breadth across harness complexities. *See the caution in §14.4.*
HIL-Ceiling(m)	$\max_k HLIS(m, HG_k)$.	Best score a standardized harness realizes from the model.
Harness Gain(m)	$\max_k HLIS(m, HG_k) - HLIS(m, HG0)$.	Capability unlocked by harnessing beyond the minimal baseline.
Harness Design Score	Mean independently verified gain when the LLM proposes harness improvements under a fixed design budget.	Ability to improve the architecture that harnesses it.
$\rho_V(m)$ **(new in 1.4)**	See §14.3.	The part of Harness Gain attributable to the model.

14.3 Verification Responsiveness ρ_V (new in 1.4)

Harness Gain is a difference of two scores. It says how much the harnessing was worth and not why, so it cannot distinguish two models with the same gain arising from different mechanisms — and those two models should be built around differently.

A harness raises two things, and only one requires the model

Acceptance and reversibility remove false completions and unrecoverable actions. They work on a model that contributes nothing beyond its first attempt.
Retry, routing and replanning require the model to convert an independently detected failure into a correction. A model that cannot do this gains nothing from them.

Definition 1 (Verification responsiveness)**.** Over runs where the acceptor rejected attempt k,

$$\rho_V = \frac{\#\{\text{accepted at } k{+}1 \text{ with no help deeper than CID1}\}}{\#\{\text{rejected at } k\}}$$

ρ_V is cheap — it needs one extra attempt on rejected runs, not a whole ladder — it is defined on the model's behavior rather than on a score difference, and it predicts which part of Harness Gain a model can realize. It is invisible to any single-attempt evaluation, because it is defined on the second attempt.

A failure mode of ρ_V, found in measurement

ρ_V computed over *coarse* criteria understates the model. In one observed case a model failed a class twice because the rejection named a rule it had not broken — the criterion bundled two rules under one name. Splitting the criterion so the message named the actual failure, and changing nothing else, moved the class from 0/2 to 2/2.

The rejection must be accurate about *what* failed, not only *that* something did. ρ_V therefore measures the pair's feedback quality as well as the model's responsiveness, and criteria granularity must be reported with it.

14.4 A Single HIL Score, and Two Cautions

For applications requiring one 0–100 number:

$$\text{HIL-Score} = 0.55 \times \text{HIL-AUC} + 0.35 \times \text{HIL-Ceiling} + 0.10 \times \text{Harnessability} \tag{13}$$

where Harnessability is a 0–100 normalization of positive Harness Gain. These weights are a proposal rather than a law. The three components and HIL-Level must always be published beside the composite.

Two structural cautions (new in 1.4)

HIL-AUC and HIL-Ceiling are computed from the same series. Ceiling is the maximum of the values AUC averages, so their correlation is structural rather than empirical, and $0.55 + 0.35 = 0.90$ of the weight sits on two views of one quantity. The only component not derivable from the others carries 0.10.

HIL-AUC does not measure harnessability. Consider a flat model scoring 55 at every rung and a steep one scoring $20, 35, 55, 75, 92$. Their AUCs are 55.0 and 55.4 — essentially equal — though only one converts architectural support into verified intelligence, which is the property §14.1 says HIL exists to measure. The composite does separate them, through the two terms carrying least weight.

Until calibration settles this, **the curve itself is the primary result** and the composite is a convenience. Report $\langle$HIL-Level, $HLIS(m, HG0)$, HIL-Ceiling, Harness Gain, ladder id$\rangle$ — the ordinal statement plus the curve's start, top and rise.

14.5 Recommended LLM Scorecard

15 Benchmark Library for Harness-Level Intelligence

A harness intelligence program should maintain a versioned benchmark registry rather than one monolithic test set. The minimal unit is a task manifest containing target coordinate/level, frozen model and harness hashes, tool/resource/authority budget, intervention policy, hidden verifier, success threshold and required retention suites.

15.1 Specification Status Must Be Recorded Per Task (new in 1.4)

A registry of task *specifications* is not a benchmark, and the difference must be visible per row rather than in aggregate. Version 1.4 requires three statuses:

- **bound** — a runnable instance exists for this band *and* this family;
- **band-only** — something runs at this difficulty band but not for this family;

Score	Subject being measured	Primary use
C-Score / model baseline	Base LLM in a minimal standardized harness.	Raw cognitive problem-solving comparison.
Delegation Surface / DI	One LLM-harness pair.	How hard a task can be delegated at each intervention budget.
Unified level U^*	One pair or organization.	Highest cumulative stage actually certified.
HLIS	One frozen pair.	Continuous comparison within and across levels.
HIL-Level + curve	One base LLM across the ladder.	How much intelligence the model expresses when harnessed.
ρ_V	The model, measured through a pair.	Whether the model converts detected failure into correction.
HDS / HDG	LLM as harness designer under frozen external evaluation.	Ability to design architectures that improve realized intelligence.

- **specification-only** — nothing runs it.

Band-only must not be counted as coverage. Substituting a software-engineering task for a document-workflow task at the same band measures a different task, and counting it reports coverage the suite does not have.

The reference library at the time of writing binds 34 of 224 specified benchmarks, marks 30 band-only and leaves 160 as specifications — a state worth reporting plainly, because a library that claims it can run something it cannot is worse than one that claims nothing.

16 Relationship to Contemporary Model and Agent Benchmarks

16.1 Existing Benchmarks Measure Important Slices

Modern model evaluation is intentionally plural: HLE [20] and MMLU-Pro [24] for broad academic reasoning; GPQA [21] for graduate-level science; FrontierMath [6] for advanced mathematics; LiveCodeBench [9] and SWE-bench [10] for coding and repository-level work; ARC-AGI [2] for fluid adaptation; BFCL [18] for tool calling; the τ-bench family [27] for interactive agent reliability; BrowseComp [25] and GAIA [14] for browsing and assistant behavior; MMMU [28] and MathVista [13] for multimodal reasoning; RULER [8] and LongBench [3] for long context; SimpleQA [16] for factuality; Chatbot Arena [4] for human preference; OSWorld [26] for computer use.

HIL does not replace these. Let $B(m)$ denote the vector of contemporary results. The HIL program **consumes** $B(m)$ as its evidence layer — A_C is built from it — and then asks a different question: how does the same frozen model behave when embedded in progressively stronger standardized harnesses?

16.2 Mapping Contemporary Benchmarks into the Architecture

The mappings are deliberately many-to-many, but no benchmark may certify an unrelated coordinate merely because it is difficult. Reports must name the exact benchmark version, task split, harness, resource budget and evaluation date.

Suite	Purpose	Example level-specific evidence
DLI-Bench	First-stage delegation surface.	T0 atomic; T2 multi-step persistence; T3 strategy construction; T4 long-horizon project; T5 hidden solution; T6 mission; TΩ charter cycles, each under H budgets.
U-Bench-U0…UΩ	Unified promotion and lower-level retention.	At U_n, rerun $U_0 \dots U_{n-1}$ retention plus the new gate.
I-Bench	Persistent individual evolution.	Restart continuity, learning transfer, Θ modification, Ψ improvement, discovery, lineage.
O-Bench	Collective intelligence.	Routing, persistent organization, adaptive coordination, reorganization, collective discovery, new-role creation.
SA-Bench	Evidence-grounded self-awareness.	State truth, capability limits, causal self-diagnosis, predicted self-change, self-unknowns, identity lineage.
C-Bench	Broad cognitive capability.	Multiple reasoning/domain families, to avoid mistaking one specialty for general cognition.
Circle-Bench	Substrate reach, reported orthogonally.	Substrate-specific implementation and validation loops.

What none of them currently report

An intervention axis. Each runs at one implicit budget — QA sets at H5, where the human supplied the entire decomposition; agent sets at roughly H0, where the run fails if the model is stuck. Neither reports a frontier.
A cost of being wrong. All score pass or fail, so a confident wrong answer and an honest escalation score identically. This is the same blindness Proposition 1 identifies inside this framework's own A_{DI}, and §13.9 is its repair.
A class that should be refused. Scoring is uniformly "attempt everything"; a suite in which refusing is always failing cannot measure judgment about when not to act.

16.3 The Main Added Quantity: Harness Response

For frozen m and generation HG_k, $HSC_m(k) = HLIS(m, HG_k)$. The ordered set is the **Harness Scaling Curve**. Two models can reverse order as harness complexity increases: one with a stronger minimal-harness profile may plateau early, while another with a weaker HG0 exploits memory, planning, verifier feedback and long-horizon state far more effectively. HIL-Level, HIL-AUC, HIL-Ceiling and Harness Gain summarize the curve, but **the curve itself should remain a primary result**.

16.4 The First Measured Curve (new in 1.4)

Version 1.3 specified the curve; nothing had instantiated the ladder. The reference ladder of §11.3 was built to HG3 and **two models** of the same family were run across it. Held identical at every rung and between models: the tool grant, the six task classes, the two seeds per class, the verifiers, the resource ceiling and the intervention budget (H1, governance only). Only the harness changed between rungs, and only the model changed between curves.

16.4.1 Result 1: the curve is invariant across the acceptance rung

For the stronger model, A_{DI} at HG0 and HG1 is not merely close but *identical*, while work handed back as finished and wrong goes 2/12 to 0/12 and two results are held back instead. HG1 adds persistent state, a criterion registered before the deliverable exists, and acceptance by a separate process. These are opposite outcomes for anyone delegating, and the same number. This is Proposition 1 observed rather than derived.

Scored with the Version 1.4 primitive of Equation (10) on the same episodes:

The repair changes what the acceptance rung is worth from nothing to +6.7 and +35.0 respectively. The weaker model gains more from it because it delivers more wrong work to be declined.

16.4.2 Result 2: Harness Gain separates the two models, and AUC does not

This is the comparison HIL exists to make, and it comes out as §14.4 predicts.

The first empirical support for the HIL construction

Harness Gain does what §14.2 claims: it distinguishes a model that converts architectural support into verified intelligence from one that was already near its ceiling. The stronger model had 6.7 points of headroom and used 3.3; the weaker had 21.7 and used 15.0.
And HIL-AUC ranks the two models in the opposite order from Harness Gain. A reader choosing which model to build a harness around wants the second ranking. This is the caution of §14.4 with data behind it, and it is the reason Version 1.4 makes the curve the primary result and the composite a convenience.

16.4.3 Result 3: a confound that the design must remove

For the weaker model, A_{DI} rose from 0.783 to 0.917 between HG0 and HG1. **This is not an acceptance effect.** An acceptance step cannot raise a success rate; it can only decline results already produced. Inspection of the per-band surfaces shows the difference is a single episode at T2 — the model produced a passing result on one run and a failing one on the other.

Both rungs perform exactly one attempt, so the two runs differ only by the model's own stochasticity. With twelve episodes per rung, that noise is larger than the effect Proposition 1 predicts, which is exactly zero.

Protocol consequence: HG0 and HG1 must share their executor outputs

Because HG0 and HG1 differ only in whether an acceptance step runs *after* the work, they can and should be scored from **the same executor outputs**: run the model once per (task, seed) and evaluate both rungs against those artifacts.
This paired design removes the between-rung variance entirely and makes the invariance testable at small sample sizes. The measurement reported here did not do this — each rung was an independent run — which is why the stronger model shows the invariance exactly and the weaker model's is masked. §17 now requires the paired design for any pair of rungs that do not differ in what the executor is asked to do.

16.4.4 What these curves do not show

The HG1→HG2 rise, where present, is the first rung whose mechanism requires the model to cooperate, consistent with §14.3; the stronger model rose +3.4 there and the weaker model did not rise at all, which is a two-point comparison and not a finding. The HG2→HG3 segment is flat or nearly so for both models because routing had one executor to choose between: the mechanism was present and inert, a defect of the instantiation rather than saturation. And both curves rest on twelve episodes per rung.

16.5 Why HIL Adds Information Beyond a Benchmark Average

16.6 Recommended Two-Layer Scorecard

16.7 Relationship to Prior AGI-Level Proposals

Morris et al. separate performance, generality and autonomy and argue for staged operational definitions rather than a binary AGI label [15]. Hendrycks et al. define AGI in terms of versatility and proficiency across a psychometrically grounded set of cognitive domains [7]. This framework preserves broad cognitive evidence in C but adds persistent individual evolution I, organizational evolution O, operational self-awareness SA, the explicit delegation surface (T,H,p) and a standardized harness-response characterization. HIL is a harness-centric complement to capability- and human-reference-centered definitions rather than a replacement.

17 Benchmark and Promotion Protocol

- Freeze the tested harness, model, tool set, prompts and authority profile before certification.
- Use multiple task families per T band so difficulty is not synonymous with one domain such as software.
- Run at predeclared intervention budgets H0–H5 with a written human-assistance policy.
- Use an independent verifier or sealed reference; the performing agent cannot certify its own success.
- Record cognitive interventions separately from governance approvals.
- Record Critical Intervention Depth so a single human-supplied central strategy cannot be hidden by a low intervention count.
- **Record the acceptance outcome beside the verifier outcome** (§13.12), or false completions and correct refusals are indistinguishable in the record. *(New in 1.4.)*
- **State each task's specification status — bound, band-only or specification-only — and do not count band-only as coverage.** *(New in 1.4.)*
- **Score rungs that differ only in post-hoc machinery from the same executor outputs.** Where two generations do not differ in what the executor is asked to do — as HG0 and HG1 do not — run the model once and evaluate both rungs against the same artifacts. Independent runs introduce between-rung variance that can exceed the effect being measured (§16.4.3). *(New in 1.4.)*
- **Publish the ladder identifier with any curve.** A curve is comparable only against the same instantiation of the generation labels. *(New in 1.4.)*
- Use confidence intervals and repeated tasks to estimate $S_A(T,H)$, then report the frontier $F_A(H,p)$.
- For U promotion, rerun a retention suite for all lower levels and then run the new-level suite.
- For I3+, O3+, SA4+ and U4+, require longitudinal evidence across multiple self/organizational changes.
- For T5+, use sealed novelty environments, hidden mechanisms or procedurally generated certification tasks to reduce memorization and leakage.

18 What Should Remain Outside the Unified Core

Not every important property should be renamed as another intelligence.

19 Worked Examples

19.1 Strong LLM, weak harness

$[C5, I0, O0, T2, H2, SA1]$. The model may solve difficult reasoning problems, but the instantiated agent has little persistent evolution and limited end-to-end delegation autonomy. High C does not automatically produce high U.

19.2 Moderate model, strong persistent harness

$[C3, I2, O1, T4, H1, SA2]$. Persistence, planning, tools, retries and verification allow difficult tasks to be delegated. Yet it is not self-improving in the I3 sense and its self-model is not causal.

19.3 Advanced organization with a bottleneck

$[C5, I4, O3, T4, H1, SA4] \rightarrow U3$, bottleneck O3. Several dimensions exceed level three, but the unified level remains U3 until the organization demonstrates recursive improvement while retaining lower-level capabilities.

19.4 Same LLM across a harness ladder

The measured case of §16.4: 93.3 → 93.3 → 96.7 → 96.7 under the v1.3 primitive, and 86.7 → 93.3 → 96.7 → 96.7 under the v1.4 primitive, from the same 48 episodes. The two scorings disagree about whether the acceptance rung is worth anything.

20 Implications for Intelligence Research

The framework distinguishes model capability from agent capability and both from system capability. A benchmark that tests only answers under carefully prepared prompts primarily measures C. A benchmark that tests whether the same system can accept a difficult goal, maintain state, choose strategy, use tools, recover, verify and finish with little human cognition measures the delegation surface. Longitudinal tests of learning and self-modification measure I; multi-agent role and reorganization tests measure O; grounded introspective tests measure SA.

Irreversibility, and the reason some classes have a delegation floor no verification removes, follows Perrow [19]; indexing an autonomy claim by the conditions it holds under follows SAE International [22]. Version 1.4 adds one implication that only appeared on contact with data. **A measurement that cannot distinguish an honest refusal from a confident error will, given a gradient to follow, select against the refusal.** This is not a hypothetical risk of some future optimizer: it was a property of this framework's own scoring variable, and it took building the ladder to see it. Any coordinate whose achievement variable is defined on "how often did it succeed" rather than "what did it deliver" should be examined for the same defect.

21 Limitations and Open Questions

- The proposed level thresholds are hypotheses and require empirical calibration across domains and models.
- Task difficulty T is multidimensional; a scalar band is operationally useful but should retain the underlying vector for horizon, uncertainty, ambiguity, verification burden, novelty and environmental change.

- Cognitive Intelligence C may require a broad psychometric profile rather than one benchmark family.
- Self-awareness is operational and evidence-grounded; the framework does not address phenomenal consciousness.
- O-levels are meaningful only for organizations with real state, authority and evidence boundaries, not a single prompt role-playing several agents.
- Open-ended levels require long horizons and are especially vulnerable to contamination, resource confounds and vague success criteria.
- Higher U denotes broader validated capability, not moral value, desirability, safety, speed or economic efficiency.

Limitations specific to the Version 1.4 measurement

Two models from one family. Both curves come from the same vendor and CLI, so the comparison controls the harness well and the model space badly. Nothing here speaks to models of other families.
Between-rung variance was not controlled. Each rung was an independent run, so the weaker model's HG0→HG1 movement is sampling noise rather than signal (§16.4.3). The paired design is now required by the protocol and was not used to produce these numbers.
Two seeds per class, twelve episodes per rung. Readings, not rates: the difference between 0.933 and 0.967 is one episode.
Four rungs of eight. HG4–HGΩ are not built, so HIL-Level above 3 is not assessable and HIL-AUC is a mean over a truncated ladder — which makes the measured composite incomparable with one computed over a full ladder.
One intervention budget. All runs at H1. A one-column surface cannot show the frontier moving leftward, and $F_A(H0, p)$ and $F_A(H2, p)$ are unmeasured.
HG3's mechanism was inert. Routing with a single executor. The flat segment is a defect of the instantiation.
ρ was fixed at 1 in the worked repair. Using each class's own loss ratio changes the net magnitudes; it does not change the direction.
Only DI was instrumented. Four of five dimensions have no runnable suite, so every measured figure is $HLIS_{DI}$ rather than HLIS.

22 Conclusion

A unified intelligence level is useful only if it summarizes rather than erases structure. This paper proposes five conceptual families — Cognitive, Individual, Organizational, Delegation and Self-Awareness — while decomposing Delegation Intelligence into two directly measured coordinates conditioned on verified reliability. The scientific profile is $[C, I, O, T, H, SA]$; the theoretical presentation remains $[C, I, O, DI, SA]$.

Unified Intelligence U0–UΩ is a cumulative gated hierarchy: promotion requires retention of all lower-level capabilities and satisfaction of minimum requirements in every coordinate. HLIS scores one frozen pair; HIL characterizes a base model across a standardized ladder; and the Harness Scaling Curve is the primary result from which the summaries are drawn.

Version 1.4 differs from its predecessors in one respect that matters more than its length. The ladder was built and a curve was measured, and the measurement found a defect in the framework's own scoring variable rather than in any system under test: A_{DI}, defined on the probability of success, was exactly invariant to the harness generation that decides whether a success may be claimed at all. The repair is small — define the delegation primitive on delivered outcomes and price false completions at the loss ratio the class already carries — and it is stated here with the data that forced it.

For harness-based AI the framework still provides a development roadmap: begin with ephemeral cognition, add persistence, learning, governed self-improvement, recursive improvement, discovery, organization, mission autonomy and finally open-ended transduction. The LLM remains the generative engine; the harness determines which loops are structurally possible; external benchmark evidence determines which level is earned. What Version 1.4 adds is the reminder that the evidence has to be able to see the thing it is measuring.

Reproducibility

The measurement in §16.4 used a deterministic task generator seeded per instance, an external verifier held outside the executor's filesystem reach, and a fixed intervention policy. The ladder identifier, per-rung episode counts, per-band success rates, false-completion counts and attempt counts are reported in-line so the reported A_{DI} can be recomputed by hand from the tables. The scoring utility reproduces the Version 1.3 figure under a $\rho = 0$ flag, so the difference between the two primitives is auditable rather than asserted.

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A Compact Reporting Template

Core measured profile: $[C, I, O, T, H, SA]$ at reliability p
Conceptual profile: $[C, I, O, DI, SA]$, where $DI = F_A(H, p)$
Unified headline: $U_n\ [C_x, I_y, O_z, T_k/H_h, SA_s]$
Extended system profile: $[U;\ C, I, O, DI, SA;\ \lambda;\ D, A, R, V, S;\ \text{resources},\ \varphi]$

B Suggested Unified Promotion Tests

C Harness-Intelligence Reporting Template

- **Base LLM:** model id / version / hash.
- **Ladder tested:** identifier, and the generations included.
- **Per harness:** U^* | HLIS | $[C, I, O, T/H, SA]$ | reliability | bottleneck | false-completion rate | resource budget.
- **LLM summary:** benchmark vector $B(m)$ | Harness Scaling Curve | HIL-Level | HIL-Ceiling | Harness Gain | ρ_V | HIL-AUC and composite, if reported, with the cautions of §14.4 | HDS if harness-design tasks were run.
- **Application panel:** selected coordinates, reported separately from the common core.

D Version History

Framework status. Level names and numerical thresholds remain proposed working definitions and should be revised when benchmark evidence warrants. Version 1.4 is the first version in which any part of the framework has been revised *because* of benchmark evidence rather than argument, and it is a revision to the measuring instrument rather than to the thing measured.

Capability slice	Representative families	Primary HIL evidence	What the benchmark alone does not establish
Broad reasoning	HLE; MMLU-Pro	A_C breadth; hard closed-ended reasoning	Persistence, autonomy, organization, self-improvement, harness scaling.
Scientific reasoning	GPQA Diamond	A_C science	Longitudinal scientific learning or autonomous research execution.
Advanced mathematics	AIME-style; FrontierMath	A_C math	Project autonomy, tool orchestration, learning across episodes.
Coding	LiveCodeBench	A_C code plus bounded execution evidence	Repository-scale planning, persistence, intervention, organization.
Software engineering	SWE-bench and repo-level suites	DI at T2–T4; execution/verifier evidence	A general level unless intervention and cross-domain retention are controlled.
Abstract / adaptive	ARC-AGI-2/3	A_C fluid; DI exploration	Persistent individual or organizational evolution.
Tool calling	BFCL	DI tool-selection layer; abstention evidence	Long-horizon mission autonomy.
Interactive agent reliability	τ-bench family	DI success surface; reliability p	General cognition or self-improvement depth.
Research / browser agent	BrowseComp; GAIA	DI for research tasks; tool reach	Longitudinal learning, organizational evolution, self-model depth.
Vision + reasoning	MMMU; MathVista	A_C multimodal	Autonomous action, persistence, harness scaling.
Long context	RULER; LongBench	A_C context handling	Whether a persistent-memory harness uses state correctly over time.
Factuality	SimpleQA	A_C factuality/calibration	Autonomy, organization, self-improvement.
Real-user preference	Chatbot Arena	External preference evidence	A mechanistic or cumulative intelligence level.
Computer agent	OSWorld family	DI in executable GUI environments	Longitudinal I/O/SA maturity unless separately instrumented.

Model	Rung	A_{DI} (v1.3)	$HLIS_{DI}$	False-done	Held back	Attempts
Stronger	HG0	0.933	93.3	**2**	0	12
	HG1	0.933	93.3	0	2	12
	HG2	0.967	96.7	0	1	15
	HG3	0.967	96.7	0	1	14
Weaker	HG0	0.783	78.3	**4**	0	12
	HG1	0.917	91.7	0	3	12
	HG2	0.917	91.7	0	3	18
	HG3	0.933	93.3	0	2	14

Table 2: Two Harness Scaling Curves on ladder `dli-ladder/HG0-HG3`. 48 episodes per model, all runs at H1. "Stronger" and "weaker" denote two members of one model family; absolute values are not a claim about any product.

	Stronger model		Weaker model	
Rung	v1.3 (ρ=0)	v1.4 net	v1.3 (ρ=0)	v1.4 net
HG0	93.3	**86.7**	78.3	**56.7**
HG1	93.3	93.3	91.7	91.7
HG2	96.7	96.7	91.7	91.7
HG3	96.7	96.7	93.3	93.3
Harness Gain	3.3	**10.0**	15.0	**36.6**

	Stronger	Weaker	Reading
$HLIS_{DI}(HG0)$	93.3	78.3	the stronger model starts far ahead
HIL-Ceiling	**96.7**	93.3	and still ends ahead
Harness Gain	3.3	**15.0**	but the weaker model is **4.5× more harnessable**
HIL-AUC	**95.0**	88.8	AUC ranks them by baseline, not by harness response

Property	Typical benchmark score	HIL extension
Unit of analysis	Model or agent in one configuration.	Frozen pair $A(m,h)$, plus the same model across a ladder.
Autonomy	Implicit or benchmark-specific.	Explicit $T \times H \times p$ surface and frontier.
Human dependence	Not a first-class variable.	H0–H5 logged and constrained.
Cost of being wrong	Absent; pass or fail.	False completions priced at ρ (§13.9).
Persistence / learning	One episode, fixed state.	I-level tests require restart continuity and learning transfer.
Organization	Fixed agent architecture.	O-level tests evaluate coordination and reorganization with real role boundaries.
Self-awareness	Usually absent.	SA requires grounded state, limitation, causal-failure and lineage tests.
Model versus harness	Conflated in the final score.	HLIS measures a pair; Harness Gain and ρ_V separate model from harness unlock.

Layer	Report	Purpose
Contemporary benchmark layer	HLE/MMLU-Pro; GPQA; math; coding/SWE; ARC-AGI; BFCL; agent/research; multimodal; long context; factuality; preference; computer use — with exact versions.	Familiar capability slices; comparison with existing leaderboards.
HIL layer	A_C profile; $S_A(T,H)$ and $F_A(H,p)$; false-completion rate; I/O/SA levels; U^*; per-harness HLIS; the curve; HIL-Level; Ceiling; Gain; ρ_V; optional composite.	How the model becomes a persistent, delegated, organizational, self-modeling system as harness sophistication increases.

Coordinate	Why it remains outside core intelligence
Circle / λ	Measures which substrate improvement loops are reached or closed; substrate access is not cognition.
Write depth D	Measures which persistent cognitive states can be rewritten; permission to rewrite does not prove intelligent rewriting.
Authority A	Permission is not intelligence. A highly intelligent verifier may deliberately have almost no action authority.
Reach R	Access to files, tools, agents, robots, labs or networks increases possible action but is not cognition.
Verification V	Determines credibility of claims; it validates intelligence rather than constituting it.
Separation S	Organizational validity condition for proposer/evaluator/promoter roles.
Resources	Compute, memory, money, time and energy affect achievable performance but must not inflate the definition.
Physical floors φ	Irreducible latency and physics constraints rather than cognitive level.

Field	Example
Unified headline	U3
Core profile	C5, I4, O3, T4/H1, SA4
Reliability	$p = 0.80$
Delegation frontier	$F_A(H0,.80) = T3$; $F_A(H1,.80) = T4$
False-completion rate *(new in 1.4)*	0.00 at H1
Bottleneck	O3
Circle reach	$\lambda = (.42,.08,.03,0,0)$
Structural profile	D3, A2, V4, separated proposer/evaluator/promoter
Resource profile	declared compute, time, memory, energy/cost budget

Transition	Minimum new evidence in addition to full lower-level retention
U0 → U1	Persistent state and correct goal-level completion after restart or context discontinuity.
U1 → U2	Verified experience improves future performance on held-out related tasks; multi-step delegation under ≤H2.
U2 → U3	Governed self/organizational method improvement plus T3 strategy construction under ≤H1 and causal self-diagnosis.
U3 → U4	Validated improvement of the improvement process plus long-horizon T4 work and accurate modeling of self-change.
U4 → U5	Independent discovery of initially unknown solution method or knowledge, and self-directed experiments about world and self-unknowns.
U5 → U6	Mission → projects → tasks → execution over changing conditions with a persistent role/mission self-model.
U6 → UΩ	Repeated bounded-charter mission generation whose validated discoveries create new tools, agents or organizations that expand future reachable problems.

Version	Date	Principal change
1.0	Aug 2026	Five families, six coordinates, the gated U0–UΩ scale, HG0–HGΩ.
1.1	Aug 2026	HLIS for a pair; HIL for a base model across a ladder; the LLM scorecard.
1.2	Aug 2026	The Harness Scaling Curve; the contemporary-benchmark mapping.
1.3	Aug 2026	Formal HLIS: symbol definitions, achievement construction, the cumulative $q^* = \min$, weights, intervals, resource envelopes, and omit-don't-zero for N/A dimensions.
1.4	Aug 2026	**The ladder built and a curve measured.** A_{DI} redefined on delivered outcomes net of false completions (§13.9, Prop. 1); the run log must record acceptance (§13.12); ρ_V added (§14.3); two structural cautions on the composite (§14.4); the reference ladder recorded (§11.3); per-task specification status required; the self-certification rule for harness design stated explicitly; first measured curve and its limitations (§16.4).